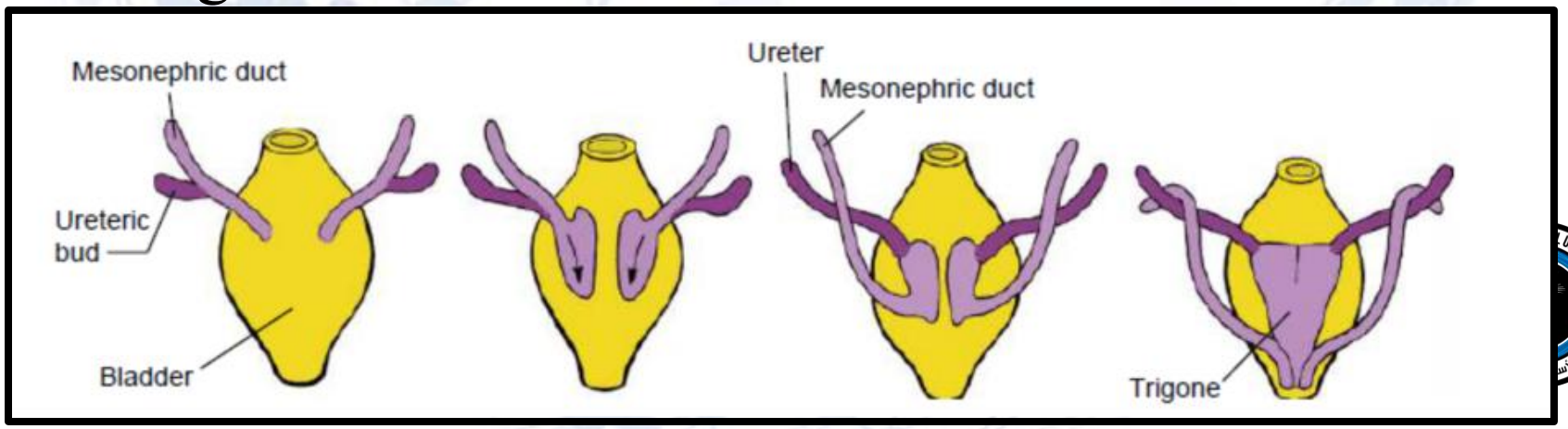
Development of Urinary Bladder

-As the bladder enlarges, distal parts of the **mesonephric ducts** are incorporated into its dorsal wall. These ducts contribute to the formation of the **trigone of the bladder**.

-As these ducts are absorbed, the ureters open separately into the urinary bladder.

- Partly because of traction exerted by the kidneys as they ascend, the orifices of the ureters move superolaterally and enter obliquely through the base of the bladder; those of the mesonephric ducts move close together to enter the prostatic urethra and in the male become the ejaculatory ducts. In females, the distal ends of the mesonephric ducts degenerate



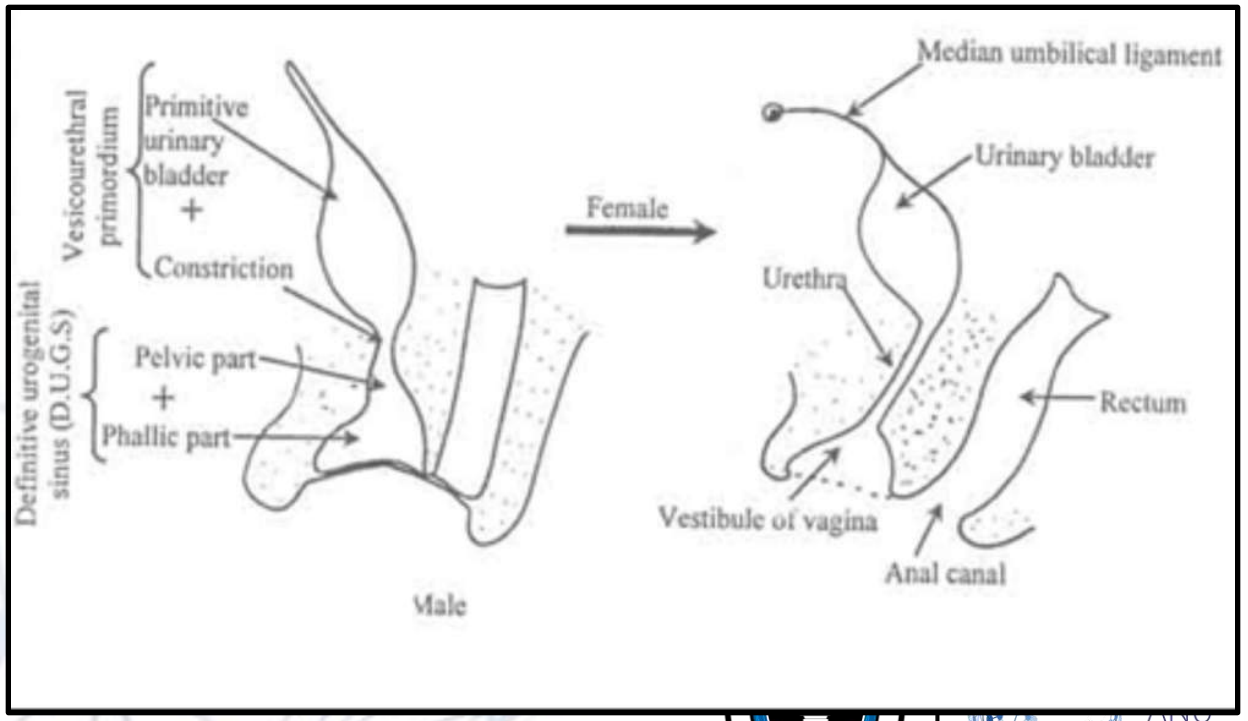
ANU MEDICAL SCHOOL PROGRAM

Development of the urethra

Male urethra:

- The **pelvic part** of the urogenital sinus will give the prostatic and membranous parts of the urethra.

- The **phallic part** of the urogenital sinus invaginates the ventral aspect of the penis and gives the penile urethra.

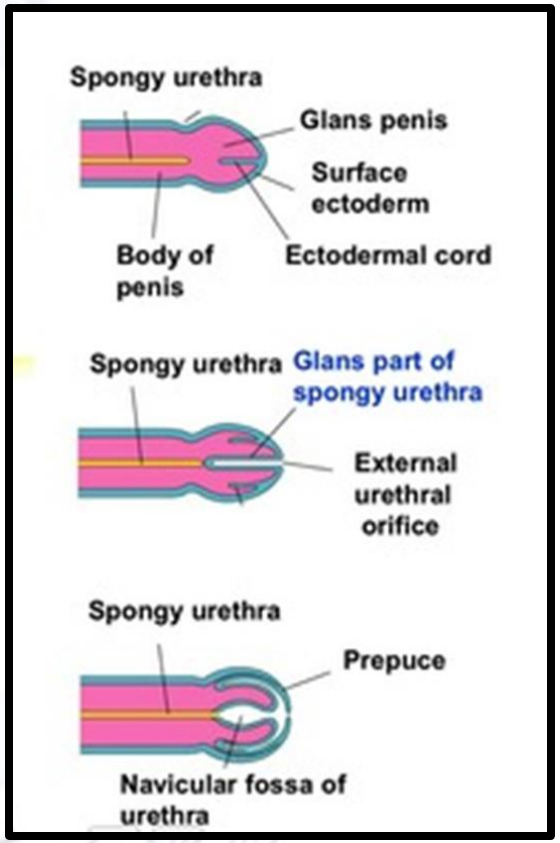


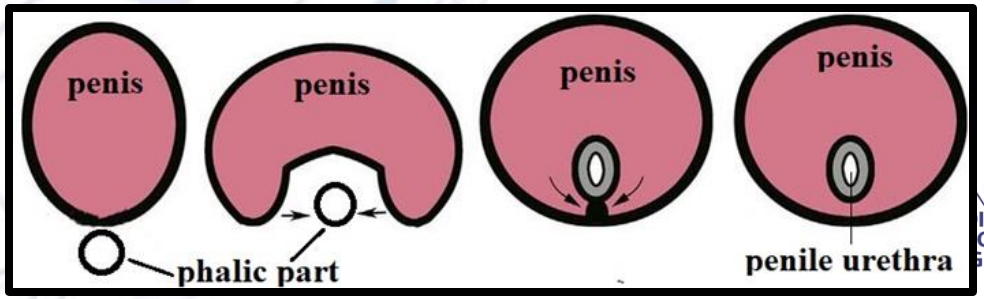
MEDICAL SCHOOL PROGRAM

Development of the urethra

- The distal part of the urethra in the glans penis is derived from a solid cord of ectodermal cells, which grows inward from the tip of the glans penis and joins the rest of the spongy urethra. So, the epithelium of the terminal part of the urethra is derived from surface ectoderm.

- The connective tissue and smooth muscle of the urethra in both sexes are derived from splanchnic mesenchyme.



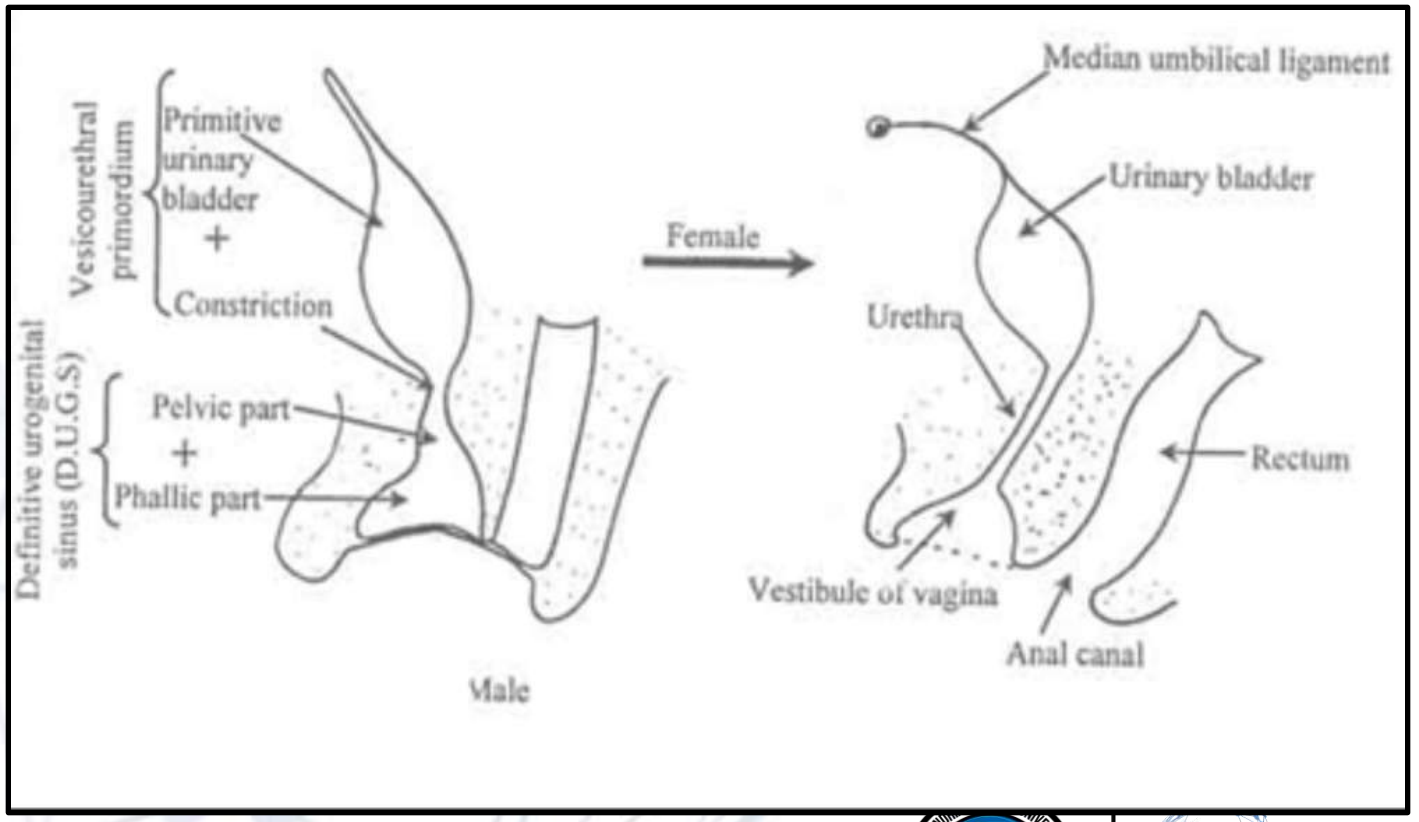


Development of the urethra

Female urethra:

-The **pelvic part** of the urogenital sinus will give the entire urethra and a part of the vagina.

- **The phallic part** will give the vestibule of the vagina.



ANU MEDICAL SCHOOL PROGRAM